



LEARN LENS

From Mistakes to Mastery

AI-Powered Adaptive Learning & Learning Verification System



Concept-Level
Assessment

Verify
Understanding



Personalized
Education

Prasunethon 2.0 — Round 2 Project Submission

Team: Tech Nova

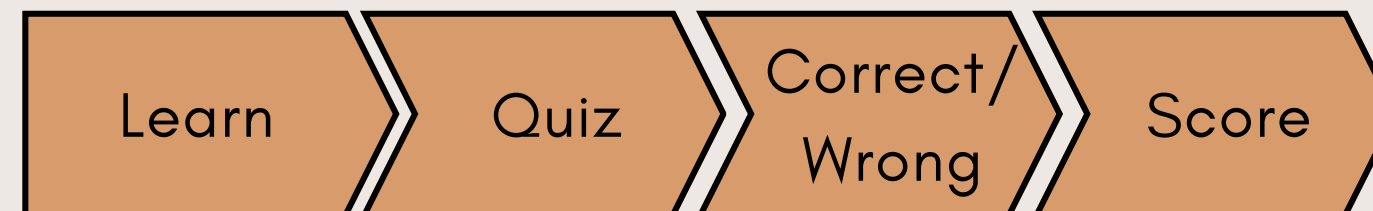
Member: Maria Dorthy

PROBLEM

Getting an Answer Right $\neq$ Understanding the Concept

Students can complete questions and still have underlying weaknesses.

Current learning systems often follow:



Core Problem

Current systems
measure answers, but
miss continuous
understanding.

Missing

- Cause of the mistake
- Repeated mistakes
- Concept understanding
- Independent problem-solving

INSIGHT

Learning Should Be Verified, Not Assumed

A learner shouldn't move forward simply because they completed a lesson.

**LearnLens evaluates
understanding through:**

Recall → Explain → Predict → Implement → Debug → Apply

Existing Approach

Lesson → Quiz → Score → Next

**LearnLens
Approach**

Learn → Attempt → Collect Evidence → Analyze → Verify → Adapt

Key Insight

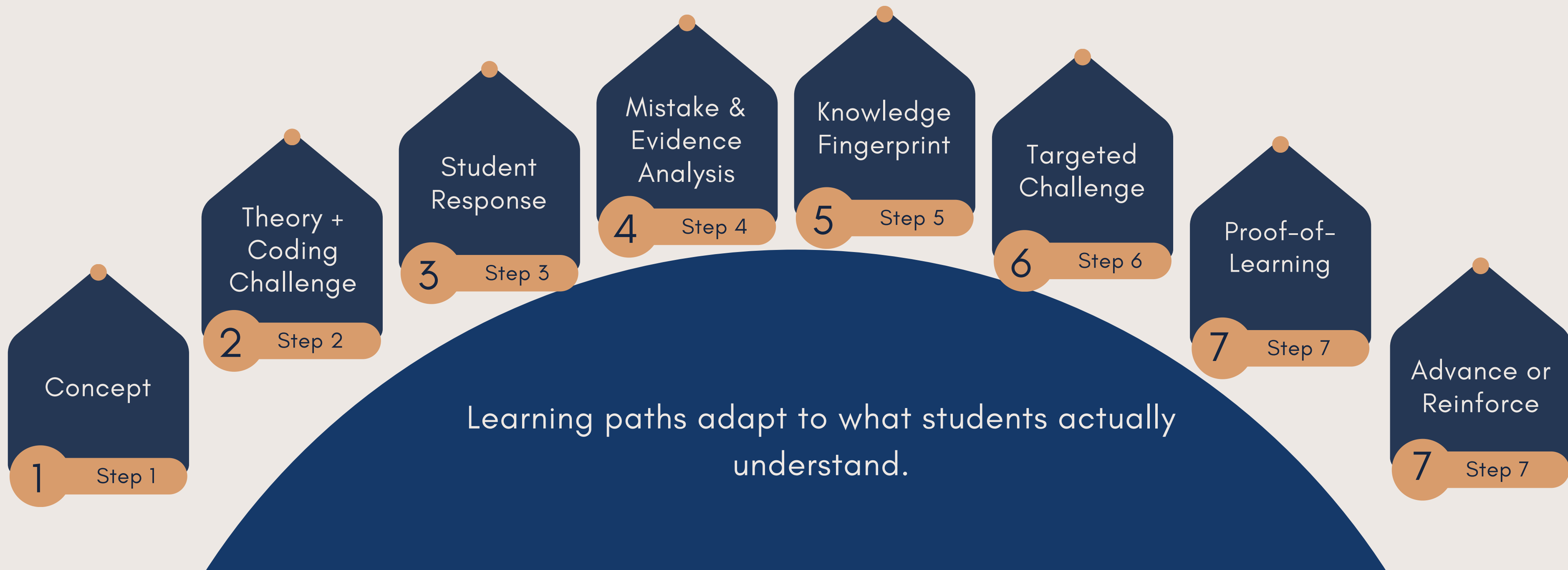
A correct answer is only one signal of
understanding.

Understanding,
backed by evidence.

THE SOLUTION

LearnLens — From Learning to Proof of Understanding

LearnLens builds a continuously evolving understanding profile for each learner.



HOW IT WORKS

The LearnLens Learning Loop

Every attempt becomes learning evidence

LEARN CONCEPT



THEORY + CODING ATTEMPT



CODE EXECUTION / RESPONSE ANALYSIS



MISTAKE DETECTION



KNOWLEDGE FINGERPRINT



TARGETED CHALLENGE



PROOF-OF-LEARN



ADAPT NEXT CONCEPT

CORE FEATURES

From Learning Activity to Learning Intelligence

01

Adaptive Learning
Roadmap

Basic → Intermediate →
Advanced, with progression
based on demonstrated
understanding.

02

Code Execution &
Mistake Tracking

Record execution errors, wrong
outputs, attempts and recurring
patterns.

03

Proof-of-Learn

Require evidence of
understanding before
considering a concept
mastered.

04

Theory + Coding
Assessment

Explain, predict, implement,
modify and debug concepts.

05

Knowledge Fingerprint

Profile understanding across
Recall, Explain, Predict,
Implement, Debug and Apply.

06

Adaptive Challenges

Give targeted challenges based
on the learner's specific
weakness.

ADAPTIVE LEARNING ROADMAP

The Roadmap Adapts to the Learner

Example

Variables → Conditions → Loops → Functions → Arrays → OOP

But progression is not based only on completion.

Example

Student A
Loops → Strong
Functions → Weak
↓
Receives additional Functions practice
↓
Reassessment
↓
Moves forward

The roadmap
changes
according to
demonstrated
understanding.

THEORY + CODING + MISTAKE INTELLIGENCE

ONE CONCEPT. MULTIPLE FORMS OF EVIDENCE.

For every concept, LearnLens can assess:

Theory

Explain → Predict → Why?

Coding

Write → Modify → Debug → Solve

Mistake Intelligence

The system records:
Error → Type → Concept
→ Recurrence →
Improvement

Example

Loop boundary error
detected 3 times



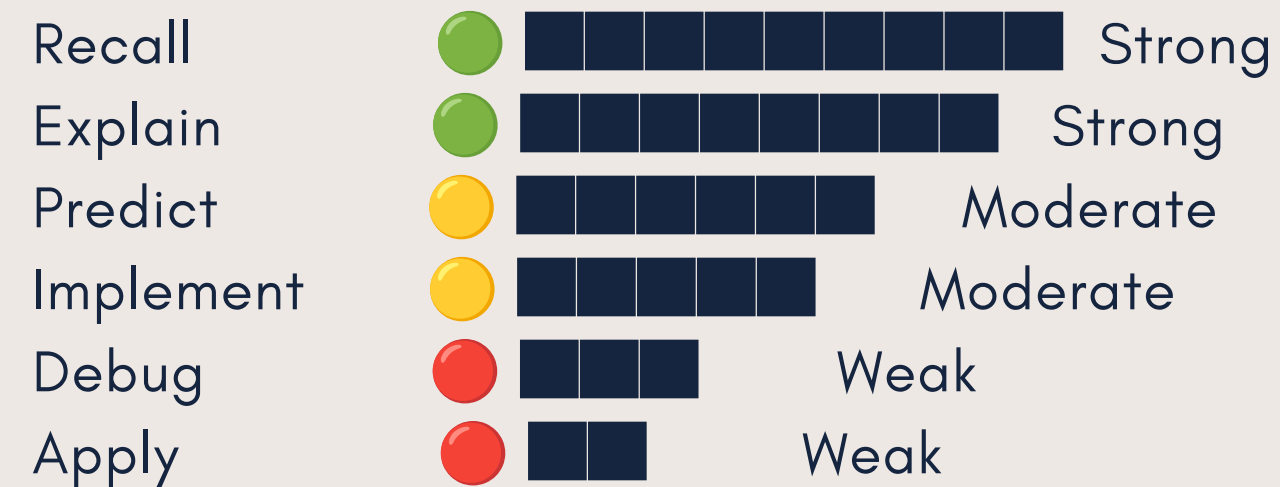
Likely weakness: boundary-
condition reasoning



Targeted practice

KNOWLEDGE FINGERPRINT

- 1 Measure How the Learner Understands
Instead of:
Recursion = 70%
- 2 LearnLens builds a Knowledge Fingerprint:
SKILL PROFICIENCY



Progress

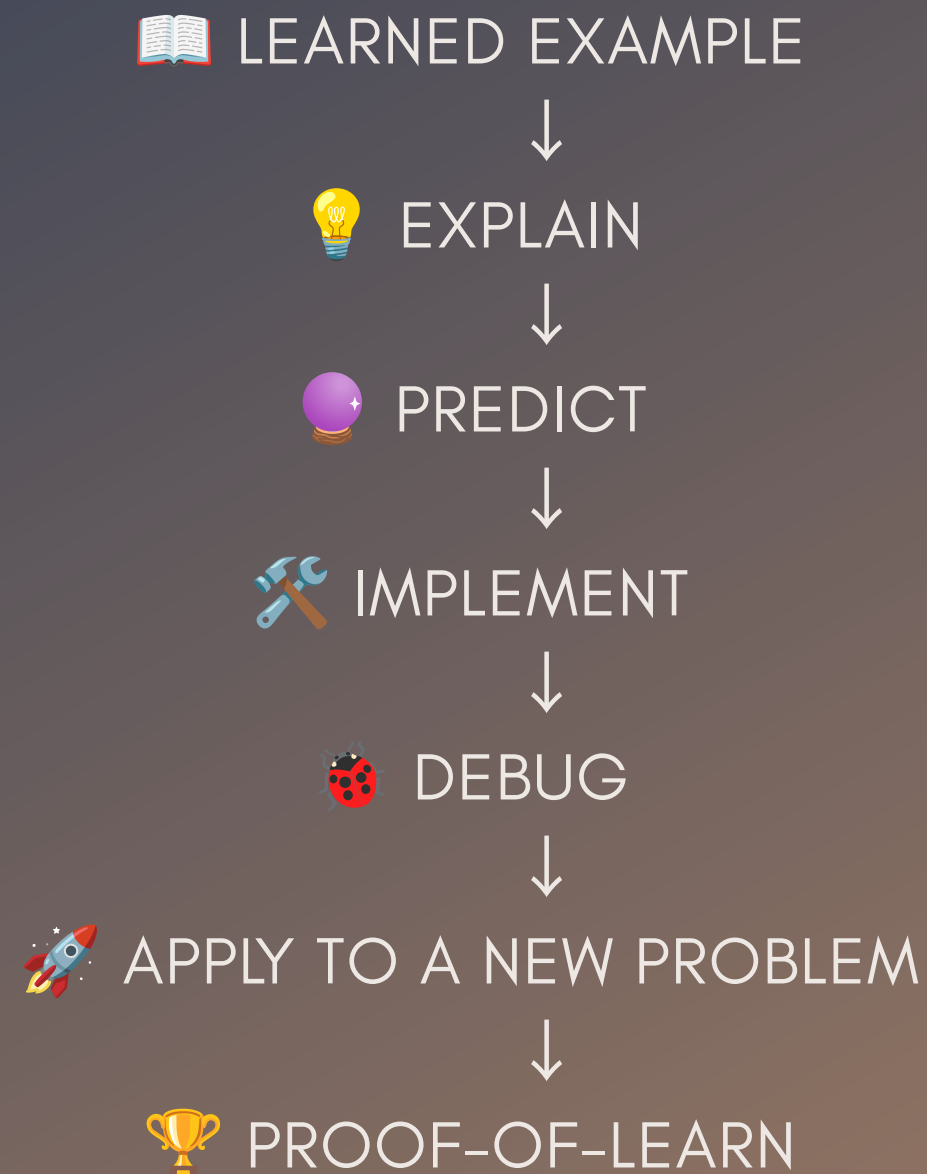
● Strong ● Moderate ● Weak

- 3 Result
The learner understands recursion theory, but struggles to debug and apply it to unfamiliar problems.

PROOF-OF-LEARN

DON'T ASSUME MASTERY. VERIFY IT.

After learning a concept, LearnLens creates a new challenge.



If understanding is sufficient:

Concept Mastered → Next Concept

If understanding is insufficient:

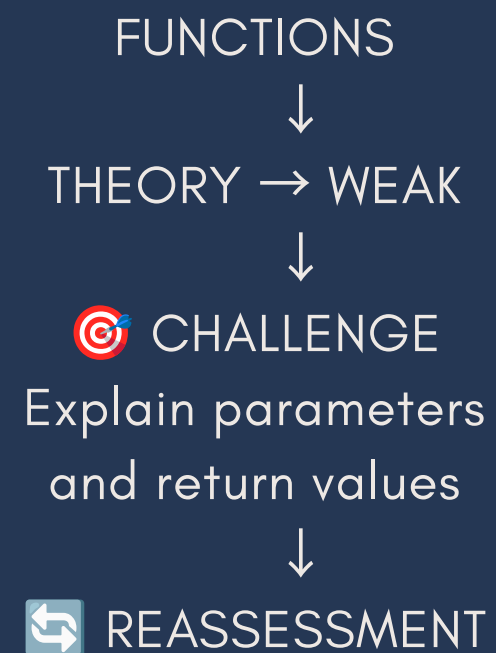
Weakness → Targeted Reinforcement

Learning is considered demonstrated only when the learner can apply the concept beyond the original example.

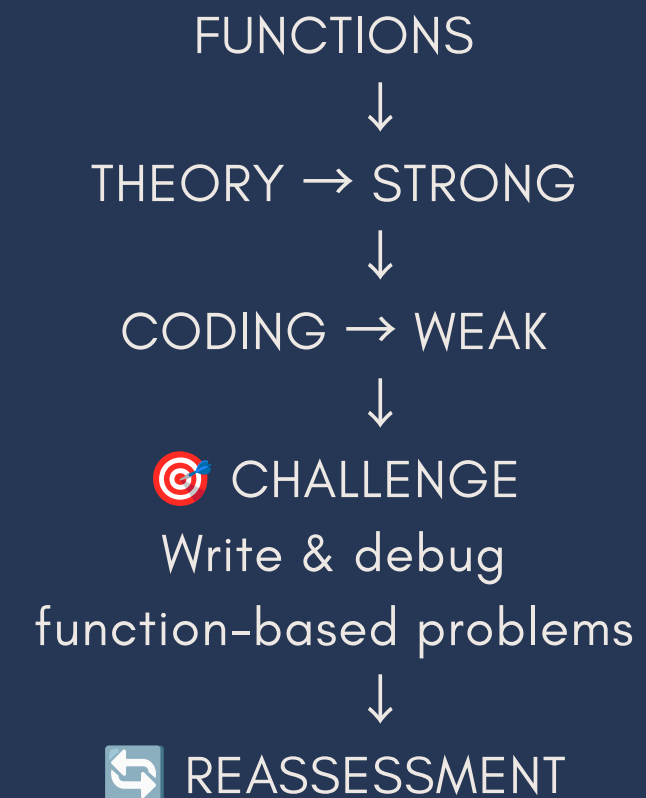
ADAPTIVE CHALLENGES

SAME TOPIC. DIFFERENT LEARNING PATH.

STUDENT A



STUDENT B



Different weakness



Different challenge

CORE INNOVATION

From “What Did You Score?” to “What Can You Actually Do?”

Traditional learning

Recursion → 70%

LearnLens

RECURSION

Dimension	Understanding
-----------	---------------

- | | | |
|-------------|---|----------|
| • Recall | → | Strong |
| • Explain | → | Strong |
| • Predict | → | Moderate |
| • Implement | → | Moderate |
| • Debug | → | Weak |
| • Apply | → | Weak |

AI-generated Insight - Result

“You understand recursion theory, but struggle to debug and apply it to unfamiliar problems.”

Innovation

Knowledge Fingerprint + Proof-of-Learn = Learning Verification

A concept is not considered mastered until the learner demonstrates understanding.

ADAPTIVE LEARNING

Same Topic. Different Learning Path.

Student A

Theory Weak
Loops
Concept Understanding → Weak
Coding → Strong
↓
Targeted Challenge
↓
Reassessment

Student B

Coding Weak
Loops
Concept Understanding → Strong
Coding → Weak
↓
Targeted Challenge
↓
Reassessment

LearnLens adapts the next challenge to the learner's actual weakness.

USER JOURNEY

One Mistake Can Change the Learning Path

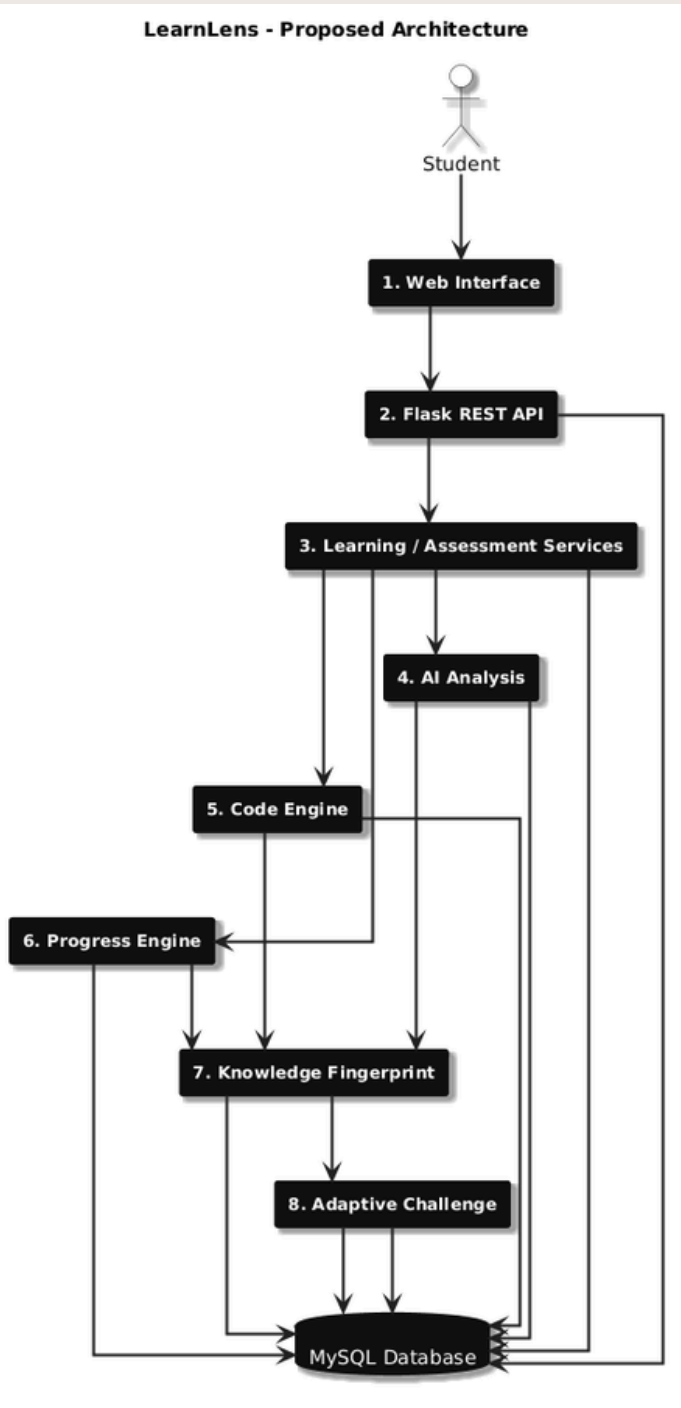


Every attempt makes the learner profile more informative.



TECHNOLOGY & ARCHITECTURE

Technically Feasible MVP Architecture



AI handles analysis and personalization.

Deterministic code handles scoring, storage, validation and progression rules.

Stack

Frontend

HTML • CSS • JavaScript

Backend

Python • Flask • REST API

Database

MySQL

AI

LLM API • Structured Response Validation

Coding Evaluation

Sandboxed Code Execution

Authentication

Password Hashing • Sessions

IMPACT & UNIQUENESS

From AI-Assisted Learning to AI-Verified Learning

Traditional LMS

- Completion
- Scores
- Progress

Generic AI Tutor

- Answers
- Explanations
- Examples

LearnLens

- Understanding Evidence
- Mistake Patterns
- Knowledge Fingerprint
- Adaptive Challenges
- Proof-of-Learn

Students

- Precise weaknesses
- Targeted practice
- Stronger independent problem-solving

Impact

Difference

Don't just measure the answer. Verify the understanding.

Educators

- Concept-level insights
- Recurring misconceptions

ROADMAP

20-23 AUGUST | BUILD → VALIDATE → DEMONSTRATE

20 AUG — FOUNDATION

- Roadmap
- Authentication
- Concepts
- Questions
- Attempts

21 AUG — EVIDENCE

- Code Execution
- Mistake Tracking
- AI Analysis

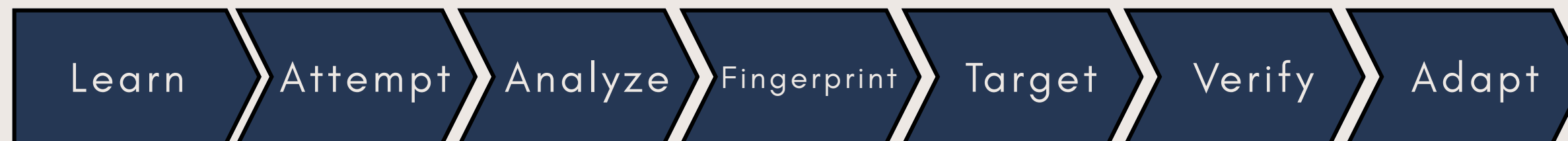
22 AUG — INNOVATION

- Knowledge Fingerprint
- Adaptive Challenges
- Proof-of-Learn

23 AUG — VALIDATION & DELIVERY

- Reassessment
- Progress
- Testing
- Deployment
- Demo

ROUND 2 MVP



Future Scope

PDF/RAG • PYQ Analysis • Teacher Analytics • Exam Priority

LEARN LENS

Every mistake becomes a learning signal.

Thank You